AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Mid-term Examination
Spring Semester 2024/2025

Date: 23 April 2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 90 minutes

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Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1- When using a StreamWriter, what happens if you omit the Close() method?.

a) Data is automatically saved
b) Data may not be written to the file
c) The file becomes read-only
d) An exception is thrown immediately

2- Which statement about RadioButton controls is FALSE?

a) They support mutual exclusion by default
b) Their Checked property is Boolean
c) They must be grouped in a Panel to work together
d) They trigger CheckedChanged events

3- What happens when you assign a decimal to an int without casting?

a) Automatic rounding occurs
b) Compilation error
c) The value is truncated
d) Overflow exception

4- What is the purpose of the TryParse() method?

a) To convert strings without throwing exceptions
b) To test mathematical expressions
c) To validate file paths
d) To compare two strings

5- Which property controls a ListBox's multi-selection capability?

a) SelectionMode
b) AllowMultiple
c) MultiSelect
d) Items.Selection

6- When you create a new project using Visual Studio, it will be stored in a _____ at the location you specify on your computer's disk.

a) Solution folder
b) Solution file
c) Project folder
d) System fileControls

7- In object-oriented programming with a GUI interface, a window displayed on the screen is called a _____.

a) Window Object
b) Form Object
c) Screen Object
d) Frame Object

8- The _____ method can be used to determine whether one string is greater than or less than another string.

a) String.Equals
b) String.IsGreater
c) String.Compare
d) String.String

9- If a program's source code contains a syntax error, it can still be translated to machine language by a compiler.

a) True
b) False

10- A variable of the decimal data type can hold real numbers with smaller precision than the double data type.

a) True
b) False

Part 2: Short Questions [20 marks]

Q1- Explain the difference between File.CreateText() and File.AppendText(). When would you use each?

Q2- Given this switch statement, Rewrite it using if-else-if with equivalent functionality.

```
switch (userRole)
{   case "Admin":
        MessageBox.Show("Full access");
        break;
    case "User":
        MessageBox.Show("Limited access");
```

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```
        break;
    default:
        MessageBox.Show("Invalid role");
        break;
}
```

Q3- What is the output of the following program? And what will be the output if we replace the sahded word with "break"?

```
int count = 0;
   while (count < 5)
   {    count++;
        MessageBox.Show(System.Convert.ToString(count));
        count++;
        if (count == 3) continue;
   }
   MessageBox.Show("Final: " + count);
```

Q4- Complete the following paragraph with the suitable terms.
[syntax error, break, exception, constant, selected, checked,const]

1- _____ is a runtime error that disrupts program flow.
2- When the application is running and it reaches a breakpoint, the application pauses and enters ---------- mode.
3- _____ is a Boolean property of a RadioButton control.
4- _____ is a keyword used to declare a constant.

Q5- Trace the following code. Specify the result of each line not only the final output.

```
using System;
namespace AdvancedTracingProblem
{  class Program
   {  static void Main(string[] args)
      {  int x = 8;
         int y = x % 3;
         int z = (int)Math.Ceiling(x / 3.0);
         Console.WriteLine("Initial Values:");
         Console.WriteLine($"x = {x}, y = {y}, z = {z}");
         for (int i = 1; i <= 3; i++)
         {  if (i == y)
            {  x *= (z + 1);
               y += (int)Math.Pow(i, 2);
            }
            else
            {  z -= x / (y + 1);
               x = (x > z) ? x - 1 : z + 1;
            }
```

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```
            Console.WriteLine($"\nAfter Iteration {i}:");
            Console.WriteLine($"x = {x}, y = {y}, z = {z}");
        }

        try
        {   Console.WriteLine("\nFinal Division:");
            Console.WriteLine(x / y);
        }
        catch (DivideByZeroException)
        {  Console.WriteLine("Cannot divide by zero!");
        }
    }
  }
}
```

Part 3: Problem Solving Questions [30 marks]

Q1- You are tasked with creating a Windows Forms application that converts temperatures between Celsius and Fahrenheit. [15 marks]

Requirements:

- The application should have a TextBox for entering temperature values.
- There should be two RadioButton controls:
  - "Celsius to Fahrenheit" (default selected)
  - "Fahrenheit to Celsius"
- There should be a "Convert" button that:
  - Performs the conversion based on the selected RadioButton
  - Displays the result in a Label with units (e.g., "100°F")
- The application must include:
  - Input validation using TryParse() to handle non-numeric input
  - A "Clear" button that resets the TextBox, Label, and RadioButtons

Formulas:

- Celsius to Fahrenheit: °F = °C × 9/5 + 32
- Fahrenheit to Celsius: °C = (°F - 32) × 5/9

Q2- Write a C# program using Windows Forms that :

- Reads numbers from a file ("numbers.txt").
- Calculates the sum of even numbers and product of odd numbers.
- Writes the results to "results.txt".
- Your program should include the following controls:
  - Button labeled "Process " to trigger the processing.

Ensure that your program handles file I/O operations correctly, calculate the sum of even numbers and the product of odd numbers accurately, and provides appropriate feedback to the user. [15 Marks]

End of questions

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